Assignment

Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

Deadline: 07-11-2026

1. Objective

The objective of this assignment is to implement and evaluate the FastSpeech 2 Text-to-Speech (TTS) model separately for English and Telugu, referring to the original research paper and an available GitHub implementation. Applicants are expected to understand the model architecture, adapt and execute the existing codebase independently for each language, train two separate models using the specified datasets, generate speech samples, and analyse the duration, pitch and energy predictions of the variance adaptor for the given test smaples.

References: FastSpeech 2 Paper | FastSpeech 2 GitHub implementation

Note: Use the GitHub implementation as a starting point and document the changes made for each language, particularly for Telugu support.

2. Datasets

The following datasets are to be used for training and validation:

Language	Dataset
English	Svarah
Telugu	IndicTTS Telugu

- Use approximately 5 hours of audio for training and 1 hour of audio for validation for each language.
- Train the English and Telugu models separately using their respective datasets.
- Specify the data selection and subsampling strategy used for each dataset.
- Ensure that the training and validation data are separate, with no overlap.
- Document the dataset preparation and preprocessing steps performed for both languages.

3. Test Samples

Generate speech for all the following test sentences using the corresponding language model. Preserve the sentence text as provided.

English Test Sentences

ID	Test sentence
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.
E2	a) The orchestra's conduct during the ceremony was praised by the audience. b) The researchers will conduct an experiment to evaluate the proposed method.
E3	The rhyth m of the orchestra changed unexpectedly.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Telugu Test Sentences

ID	Test sentence
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.
T2	a) తాడుపై అతని పట్టు చాలా బలంగా ఉంది. b) ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.
T3	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

4. Deliverables

a. Code Submission

Submit the complete implementation with the following folder structure in your GitHub repository:

```
project/
├── English/
│   ├── train.py
│   ├── preprocess.py
│   ├── inference.py
│   ├── config.yaml
│   └── checkpoints/
├── Telugu/
│   ├── train.py
│   ├── preprocess.py
│   ├── inference.py
│   ├── config.yaml
│   └── checkpoints/
├── Results/
│   ├── English/
│   │   ├── training_loss.png
│   │   ├── validation_loss.png
│   │   ├── test_generated_audios/
│   │   └── variance_adaptor_plots/
│   │       ├── duration/
│   │       ├── pitch/
│   │       └── energy/
│   └── Telugu/
│       ├── training_loss.png
│       ├── validation_loss.png
│       ├── test_generated_audios/
│       └── variance_adaptor_plots/
│           ├── duration/
│           ├── pitch/
│           └── energy/
├── Report/
│   └── Technical_Report.pdf
├── README.md
└── requirements.txt
```

Submission Requirements

1. Include complete preprocessing, training and inference code for both language models.
2. Include the modified files from the selected GitHub implementation and separate configuration files for English and Telugu.
3. Include the trained model checkpoints separately for English and Telugu.
4. Include training-loss and validation-loss plots for both models.

5. Include generated audio samples for all five test sentences in each language.
6. Include variance-adaptor plots for predicted duration, pitch (F0) and energy for each test sentence, for both languages.
7. Maintain the specified folder structure in the GitHub repository.

Technical Report

The report must include the following details and analysis:

- Implementation approach, architecture components used from the reference implementation, and modifications made for English and Telugu.
- Dataset selection, subsampling, preprocessing, train/validation split, and training configuration for each language.
- Analysis of training and validation loss curves, including convergence behaviour and any instability or overfitting observed.
- Observations on the generated speech, including pronunciation, and overall speech quality for the provided each test sentence.
- Analysis of the variance-adaptor plots for duration, pitch (F0) and energy, with explanations of the observed patterns for each of the test sentence.
- Comparative observations between the English and Telugu models, including limitations, failures and possible improvements based on the test sentences that are provided.
- Reproducibility details, including software dependencies, configuration files and commands required to preprocess, train and run inference.

b. GitHub Repository Submission

- Create a GitHub repository containing the complete project.
- Ensure that the repository is publicly accessible.
- Submit the GitHub repository URL through the assignment submission form.
- Ensure that all code, results, checkpoints and the technical report are accessible for evaluation.

Note: If model checkpoints exceed GitHub's file-size limits, use Git LFS or provide an accessible external download link in the README. Ensure that the link remains accessible during evaluation.